A REPRESENTABLE MONOMORPHISM THAT IS AN ISOMORPHISM OVER THE REDUCTION IS A CLOSED IMMERSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains the author's own wording 'in the previous parapgrah' and the abbreviation 'ie.'; both lie inside the counted span and are reproduced verbatim, as the delivered text is the author's.

Editorial note. The proof explicitly refers to a counterexample in another chapter to justify the role of the fourth assumption; that reference is carried as a navigation stub and recorded as a prerequisite.

Editorial note. The counted spans carry 7 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Closed immersions of formal algebraic spaces are etale local on the target. Same chapter, Lemma lemma-characterize-closed-immersion, invoked at proof lines 6212-6214. Allows the target to be assumed affine.
- Affine formal algebraic spaces as filtered colimits of affine schemes. Same chapter, Definition definition-affine-formal-algebraic-space, invoked at proof lines 6216-6217. Supplies the system whose base changes are studied.
- An algebraic space that is a monomorphism over an affine scheme and induces an isomorphism on reductions is an affine scheme. Limits of Spaces, Proposition spaces-limits-proposition-affine, invoked at proof lines 6220-6222. Reduces the assertion about formal algebraic spaces to an assertion about affine schemes.
- A morphism of rings is surjective if it is surjective modulo a locally nilpotent ideal, and Nakayama's lemma in the finitely generated case. Algebra, Lemma algebra-lemma-surjective-mod-locally-nilpotent at proof lines 6242-6244, and Algebra, Lemma algebra-lemma-NAK part (11) at proof lines 6245-6247. Give surjectivity of the ring map in the two cases of the case split.
- The general statement fails: a monomorphism of affine schemes inducing an isomorphism on reductions need not be a closed immersion. Examples, Section examples-section-epimorphism-not-surjective, invoked at proof lines

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

6229-6231. Explains why the fourth assumption is needed and isolates what the proof has to use.

Source reference. Original lemma 00DV: sources/stacks-chow/tags.txt, source lines 512–512. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 07RD: sources/stacks-chow/tags.txt, source lines 9863–9863. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original section 06RH: sources/stacks-chow/tags.txt, source lines 8641–8641. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0GHY: sources/stacks-etale/pinned-original-source.tex, source lines 881–881. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original proposition 07VT: sources/stacks-chow/tags.txt, source lines 10018–10018. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original remark 0AIS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2769–2769. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let S be a scheme. We say a sheaf X on $(Sch/S)_{fppf}$ is an *affine formal algebraic space* if there exist

(1) a directed set Λ,
(2) a system $(X_\lambda, f_{\lambda\mu})$ over Λ in $(Sch/S)_{fppf}$ where
   (a) each X_λ is affine,
   (b) each $f_{\lambda\mu} : X_\lambda \to X_\mu$ is a thickening,

such that

$$X \cong \operatorname{colim}_{\lambda\in\Lambda} X_\lambda$$

as fppf sheaves and X satisfies a set theoretic condition (see Remark 0AIS). A *morphism of affine formal algebraic spaces* over S is a map of sheaves.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of formal algebraic spaces. Assume*

(1) *f is representable by algebraic spaces,*
(2) *f is a monomorphism,*
(3) *the inclusion $Y_{red} \to Y$ factors through f, and*
(4) *f is locally of finite type or Y is locally Noetherian.*

Then f is a closed immersion.

Proof. Assumptions (2) and (3) imply that $X_{red} = X \times_Y Y_{red} = Y_{red}$. We will use this without further mention.

If $Y' \to Y$ is an étale morphism of formal algebraic spaces over S, then the base change $f' : X \times_Y Y' \to Y'$ satisfies conditions (1) – (4). Hence by Lemma 0GHY we may assume Y is an affine formal algebraic space.

Say $Y = \mathrm{colim}_{\lambda \in \Lambda} Y_\lambda$ as in Definition 0.1. Then $X_\lambda = X \times_Y Y_\lambda$ is an algebraic space endowed with a monomorphism $f_\lambda : X_\lambda \to Y_\lambda$ which induces an isomorphism $X_{\lambda,red} \to Y_{\lambda,red}$. Thus X_λ is an affine scheme by Limits of Spaces, Proposition 07VT (as $X_{\lambda,red} \to X_\lambda$ is surjective and integral). To finish the proof it suffices to show that $X_\lambda \to Y_\lambda$ is a closed immersion which we will do in the next paragraph.

Let $X \to Y$ be a monomorphism of affine schemes such that $X_{red} = X \times_Y Y_{red} = Y_{red}$. In general, this does not imply that $X \to Y$ is a closed immersion, see Examples, Section 06RH. However, under our assumption (4) we know that in the previous parapgrah either $X_\lambda \to Y_\lambda$ is of finite type or Y_λ is Noetherian. This means that $X \to Y$ corresponds to a ring map $R \to A$ such that $R/I \to A/IA$ is an isomorphism where $I \subset R$ is the nil radical (ie., the maximal locally nilpotent ideal of R) and either $R \to A$ is of finite type or R is Noetherian. In the first case $R \to A$ is surjective by Algebra, Lemma 07RD and in the second case I is finitely generated, hence nilpotent, hence $R \to A$ is surjective by Nakayama's lemma, see Algebra, Lemma 00DV part (11). □